

Data Analyst Simple Workflow

Employee Salary Analysis



Imagine the Boss requesting to analyze employee salary data to map out the salary distribution by department and job title, highlighting any key observations.

We need to understand our compensation structure and how it relates to employee longevity. This will help us identify potential areas of concern regarding salary fairness and employee retention.



Workflow

01

Pulling Dataset

Pull the requested data
from relational database
with SQL query

02

Analysis

Analyze the data in excel
into various metrics with
Pivot Tables

03

Visualization

Visualize the data into
readable graph and chart

04

Insight

Present the insight

Pull and Load Data

01

Pulling Data

[SQL Worksheet]*



Aa



```
1  SELECT FIRST_NAME || ' ' || LAST_NAME
2     AS FULL_NAME,
3     JOB_TITLE,
4     DEPARTMENT_NAME,
5     HIRE_DATE,
6     SALARY,
7     TRUNC(MONTHS_BETWEEN(SYSDATE, HIRE_DATE)/12)
8     AS YEARS_OF_SERVICE
9  FROM HR.EMPLOYEES E
10 INNER JOIN HR.DEPARTMENTS D
11 ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
12 INNER JOIN HR.JOBS J
13 ON E.JOB_ID = J.JOB_ID
14
```



For this simulations, im using HR
Databases on freesql.com

This step involves constructing the SQL
query based on the required analysis
criteria to extract the relevant employee
salary data.

Query result

Script output

DBMS output

Explain Plan

SQL history



Download

Execution time: 0.011 seconds

	PARTMENT_NAM	HIRE_DATE	SALARY	YEARS_OF_SERVICE
25	rchasing	11/15/2016, 12:00:00	2600	8
26	rchasing	8/10/2017, 12:00:00	2500	8
27	rchasing	12/7/2012, 12:00:00	11000	12

Load CSV Data into Excel Table

employee_salary_and_job_title.csv

File Origin	Delimiter	Data Type Detection
1252: Western European (Windows) ▾	Comma ▾	Based on first 200 rows ▾

FULL_NAME	JOB_TITLE	DEPARTMENT_NAME	HIRE_DATE	SALARY	YEARS_OF_SERVICE
William Gietz	Public Accountant	Accounting	07/06/2012 07.00.00	8300	13
Shelley Higgins	Accounting Manager	Accounting	07/06/2012 07.00.00	12008	13
Jennifer Whalen	Administration Assistant	Administration	17/09/2013 07.00.00	4400	12
Steven King	President	Executive	17/06/2013 07.00.00	24000	12
Lex Garcia	Administration Vice President	Executive	13/01/2011 07.00.00	17000	14
Neena Yang	Administration Vice President	Executive	21/09/2015 07.00.00	17000	10
John Chen	Accountant	Finance	28/09/2015 07.00.00	8200	10
Daniel Faviat	Accountant	Finance	16/08/2012 07.00.00	9000	13
Luis Popp	Accountant	Finance	07/12/2017 07.00.00	6900	7
Ismael Sciarra	Accountant	Finance	30/09/2015 07.00.00	7700	10
José Manuel Urman	Accountant	Finance	07/03/2016 07.00.00	7800	9

Since the data is already clean (missing values, unknown characters, etc.), the next step is simply to import the CSV file and structure it as an Excel table.

Analyze Data

02

Create Pivot Tables

Average Salary by Job Title		
Job Title	Average Salary	Total Employee
President	\$24.000,00	1
Administration Vice President	\$17.000,00	2
Marketing Manager	\$13.000,00	1
Sales Manager	\$12.200,00	5
Accounting Manager	\$12.008,00	1
Finance Manager	\$12.008,00	1
Purchasing Manager	\$11.000,00	1
Public Relations Representative	\$10.000,00	1
Sales Representative	\$8.396,55	29
Public Accountant	\$8.300,00	1
Accountant	\$7.920,00	5
Stock Manager	\$7.280,00	5
Human Resources Representative	\$6.500,00	1
Marketing Representative	\$6.000,00	1
Programmer	\$5.760,00	5
Administration Assistant	\$4.400,00	1
Shipping Clerk	\$3.215,00	20
Stock Clerk	\$2.785,00	20
Purchasing Clerk	\$2.780,00	5
Grand Total	\$6.456,75	106

Average Salary by Department		
Department Name	Average Salary'	Total Employee'
Executive	\$19.333,33	3
Accounting	\$10.154,00	2
Public Relations	\$10.000,00	1
Marketing	\$9.500,00	2
Sales	\$8.955,88	34
Finance	\$8.601,33	6
Human Resources	\$6.500,00	1
IT	\$5.760,00	5
Administration	\$4.400,00	1
Purchasing	\$4.150,00	6
Shipping	\$3.475,56	45
Grand Total	\$6.456,75	106

The next step is to create pivot tables to visualize the salary distribution by department and job title, and to calculate the correlation between salary and years of service.

Create Pivot Tables

Employee Salary vs Years of Services

Row Labels	Salary1	Years of Service
Steven King	\$24.000,00	12
Lex Garcia	\$17.000,00	14
Neena Yang	\$17.000,00	10
John Singh	\$14.000,00	11
Karen Partners	\$13.500,00	10
Michael Martinez	\$13.000,00	11
Nancy Gruenberg	\$12.008,00	13
Shelley Higgins	\$12.008,00	13
Alberto Errazuriz	\$12.000,00	10
Lisa Ozer	\$11.500,00	10
Den Li	\$11.000,00	12
Gerald Cambault	\$11.000,00	8
Ellen Abel	\$11.000,00	11
Eleni Zlotkey	\$10.500,00	7
Clara Vishney	\$10.500,00	10
Sean Tucker	\$10.000,00	10

Total Salary	Total Employee	Average Year of Service	Min of SALARY	Max of SALARY
\$684.416,00	106	9,47	2100	24000

Top 3 Highest Salary

Row Labels	Sum of SALARY
Steven King	24000
Lex Garcia	17000
Neena Yang	17000
Grand Total	58000

The next step is to create pivot tables to visualize the salary distribution by department and job title, and to calculate the correlation between salary and years of service.

Visualize Data

03

Create Visual Chart



I create 3 different key metrics and we can see the total salary, total employee, and the average years of service.

All displayed metrics will dynamically update based on the selection made using the Department slicer.

Average Salary by Department



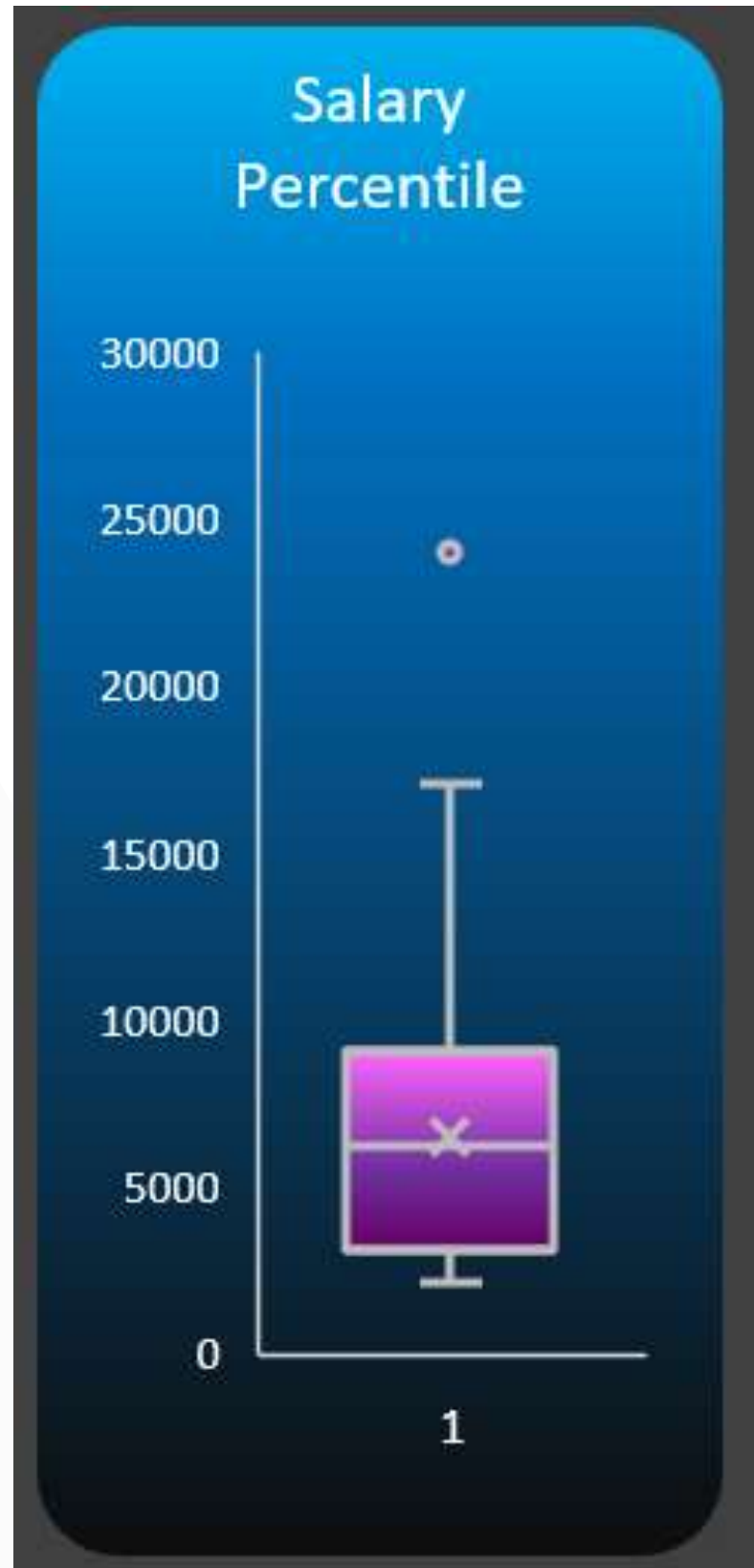
This chart/section illustrates the average salary distribution across each department. Notably, the three departments with the lowest average salaries are Shipping, Purchasing, and Administration.

Employee Salary vs Tenure



This chart illustrates the correlation between Salary and Tenure (Years of Service). The data shows a slightly positive correlation, suggesting a general trend where employees with longer tenure within the company also tend to have higher salaries.

Salary Box Plot



This box plot illustrates the salary distribution using the five-number summary: Minimum, First Quartile (25%), Median (50%), Third Quartile (75%), and Maximum. We can see one outlier, which corresponds to the President's highest salary. The majority of salaries are concentrated in the \$3,000 to \$9,000 range.

Dashboard

Shipping

Employee Salary and Job Analysis

Total Salary
\$684.416,00

Total Employee
106

Average Years of Service
9,47

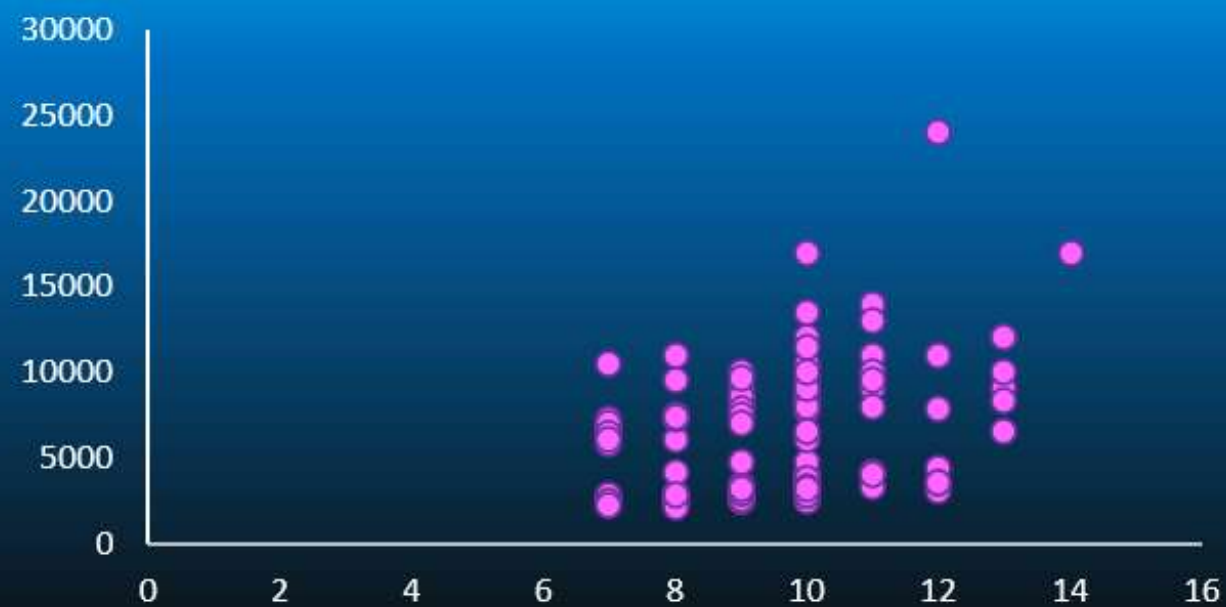
Highest Salary by Employee



Salary Percentile



Employee Salary vs Years of Service



Average Salary by Department



Insight

04



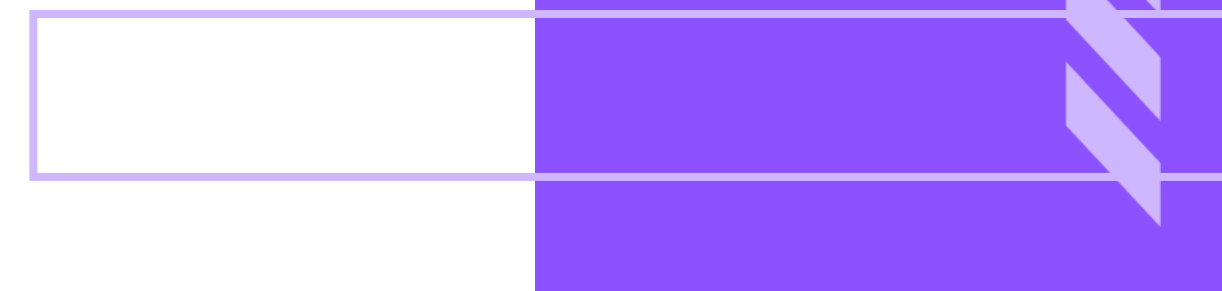
Insight conclusion

Executive Salary and Distribution

The average salary distribution is significantly wide within the Executive department (**President and Vice President**). This is normal because they are in top-tier leadership roles that carry the highest level of organizational responsibility, strategic decision-making authority, and scarcity of specialized skillsets, leading to significantly higher and more varied compensation packages compared to other roles.

Departmental Average Salaries

The average salary is significantly low in the **Shipping, Purchasing, and Administration Departments**. (This typically suggests these roles are less specialized, entry-level, or transactional in nature compared to other areas like Public Relations or Accounting).





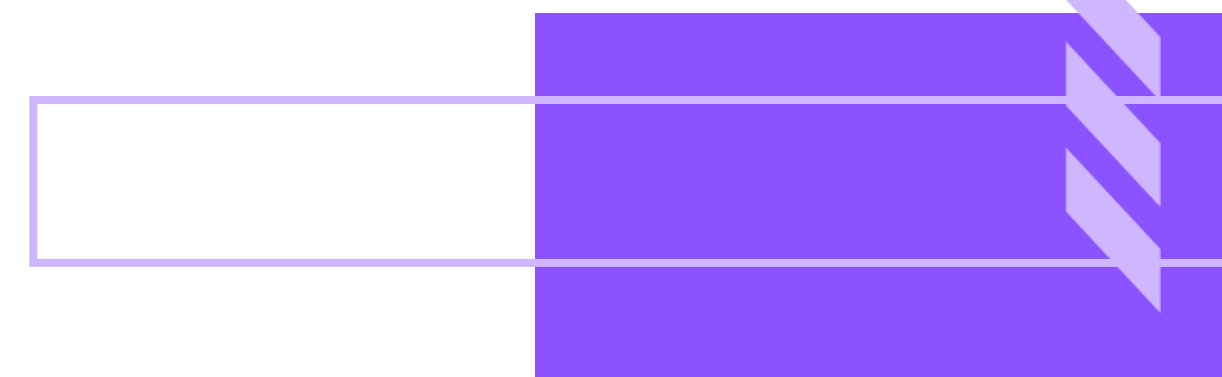
Insight conclusion

Salary-Tenure Correlation

There is a slightly positive correlation between salary and years of service (tenure), indicating that longer employee tenure is generally associated with salary growth.

Overall Salary Concentration

The overall distribution of salary is concentrated at the lower end, primarily falling within the **\$3,000 to \$9,000** range. (This suggests the majority of the workforce is composed of employees in operational or mid-level roles, or that the company operates in a lower-cost labor market).





Thank You

For further information regarding this analysis or my experience, please feel free to reach out using the following channels:

Email : yusrifargfar@gmail.com

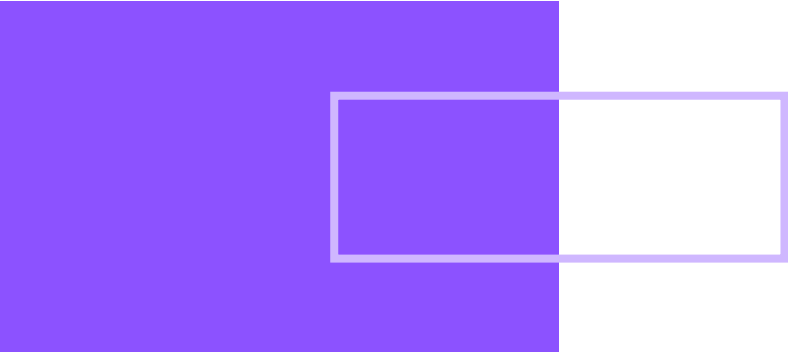
WhatsApp : +62 896 8511 3550

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LinkedIn : www.linkedin.com/in/yusrifar-gaffar-69bb38323

GitHub : www.github.com/yusrifargfar





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